



AI Build Sprint – Problem Statement 5

Audit-Ready AI Decision System (Explainable + Traceable AI)

◆ Objective

Build a no-code AI system for **insurance claim processing** that not only produces a decision, but also generates a **complete, structured audit trail**.

Your system should clearly show:

- Input
- Intermediate steps
- Final decision

Each step must be **transparent, explainable, and traceable**.

◆ System Requirements

Your system must include the following stages:

1. **Claim Analysis**
 2. **Coverage Validation**
 3. **Fraud / Consistency Check**
 4. **Decision Generation**
 5. **Audit Log Generation** (core component)
-

◆ Input Format

Claim ID: C1

Description: Hit a tree while reversing

Policy Type: Comprehensive / Third-Party

Claim Amount: ₹15,000

Past Claims: 0

◆ Decision Rules

Coverage:

- **Comprehensive Policy** → Covers own damage
 - **Third-Party Policy** → Does NOT cover own damage
-

Validity:

- Claim must be:
 - Covered under policy
 - Supported by documents
-

Fraud Indicators:

- Past Claims > 3
 - High claim for minor damage
 - Missing or inconsistent inputs
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◆ Audit Requirements (Critical)

Your system must generate a **step-by-step audit trail**, where each step includes:

- **Step name**
 - **Reasoning**
 - **Confidence score**
 - **Source or assumption**
-

◆ Audit Log Format (Mandatory)

```
[  
{  
  "step": "claim_analysis",
```

```
"reason": "Accident described is valid",
"confidence": 0.85,
"source": "Description input"
},
{
  "step": "coverage_check",
  "reason": "Third-party policy does not cover own damage",
  "confidence": 0.95,
  "source": "Policy Type rule"
},
{
  "step": "decision",
  "result": "Rejected",
  "confidence": 0.92,
  "source": "Coverage result"
}
]
```

◆ Final Output Format

```
{
  "Claim ID": "C1",
  "Status": "Rejected",
  "Reason": "Policy does not cover own damage",
  "Confidence Score": 0.92,
  "Audit Log": [ ... ]
}
```

◆ Explanation Requirement

- Each step must clearly explain:
 - What was checked
 - Why the result was obtained
 - Final decision must align with the audit log
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◆ Confidence Score

Each step and final output must include a confidence score:

- High → clear and rule-based
 - Medium → some ambiguity
 - Low → missing or conflicting data
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◆ Multiple Case Processing

Your system must process **at least 3 insurance claims** and generate audit logs for each.

◆ Consistency Requirement

- Final decision must match intermediate steps
 - No contradictions between:
 - Audit log
 - Final output
-

◆ Sample Case

Description: Hit a tree while reversing

Policy: Third-party only

Claim Amount: ₹15,000

Expected Behavior:

- Claim Analysis → Valid accident
 - Coverage Check → Not covered
 - Decision → Rejected
-

Final Output:

- Status: Rejected
 - Reason: Policy does not cover own damage
 - Confidence: High
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◆ Enhancements

- Provide more detailed reasoning for each step
 - Include additional intermediate checks (e.g., document validation, damage assessment)
 - Improve confidence scoring using multiple factors
 - Present audit log in a structured table or visual format
 - Ensure traceability from input fields to final decision
 - Handle edge cases such as missing or conflicting data
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◆ Mandatory Requirements

- Generate **step-by-step audit trail**
 - Include **reasoning, confidence, and source** for each step
 - Maintain **consistency between steps and final output**
 - Process multiple claims
 - Provide structured output (JSON or table)
-

◆ Tools

Use **no-code AI tools only**

◆ Key Instruction

Your system must make decisions that can be **clearly understood, verified, and audited**.

◆ Layman One-Liner

Build a system that not only gives a decision, but also shows **exactly how and why it made that decision**.